

# **FORMULAS FOR REFERENCE**

|          |                        |                                                              |
|----------|------------------------|--------------------------------------------------------------|
| SPHERE   | Surface area           | $= 4\pi r^2$                                                 |
|          | Volume                 | $= \frac{4}{3}\pi r^3$                                       |
| CYLINDER | Area of curved surface | $= 2\pi rh$                                                  |
|          | Volume                 | $= \pi r^2 h$                                                |
| CONE     | Area of curved surface | $= \pi rl$                                                   |
|          | Volume                 | $= \frac{1}{3}\pi r^2 h$                                     |
| PRISM    | Volume                 | $= \text{base area} \times \text{height}$                    |
| PYRAMID  | Volume                 | $= \frac{1}{3} \times \text{base area} \times \text{height}$ |

There are 36 questions in Section A and 18 questions in Section B.  
The diagrams in this paper are not necessarily drawn to scale.

## **Section A**

Label 1 If  $A = \frac{h}{2}(a+b)$ , then  $b =$

- A.  $2A - ah$
- B.  $\frac{2}{h}(A - a)$
- C.  $\frac{2A - a}{h}$
- D.  $a - \frac{2A}{h}$
- E.  $\frac{2A}{h} - a$

Label 2 Factorize  $x^2 - x - xy + y$

- A.  $(x - y)(x - 1)$
- B.  $(x - y)(x + 1)$
- C.  $(x + y)(x - 1)$
- D.  $(1 - x)(x + y)$
- E.  $(1 + x)(y - x)$

Label 3 Simplify  $\frac{(a^3b^{-1})^{-2}}{(a^{-1}b^2)^4}$

- A.  $\frac{1}{ab^3}$
- B.  $\frac{1}{a^2b^3}$
- C.  $\frac{1}{a^2b^6}$
- D.  $\frac{1}{a^2b^9}$
- E.  $\frac{a^4}{b^6}$

Label 4 Let  $f(x) = 3x^2 + ax - 7$ . If  $f(-1) = 0$ , find  $f(-2)$ .

- A.  $-27$
- B.  $-11$
- C.  $-3$
- D.  $1$
- E.  $3$

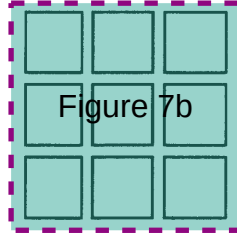
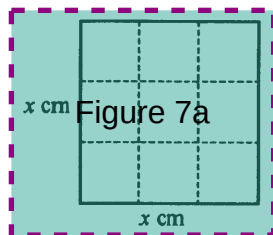
Label 5 If  $\begin{cases} y = x^2 - 1 \\ y = 2x - 2 \end{cases}$ , then  $y =$

- A.  $-4$
- B.  $0$
- C.  $1$
- D.  $0$  or  $8$
- E.  $-4$  or  $4$

Label 6 Find the values of  $x$  which satisfy both  $x + 3 > 0$  and  $-2x < 1$ .

- A.  $x > -3$
- B.  $x > -\frac{1}{2}$
- C.  $x > \frac{1}{2}$
- D.  $-3 < x < -\frac{1}{2}$
- E.  $-3 < x < \frac{1}{2}$

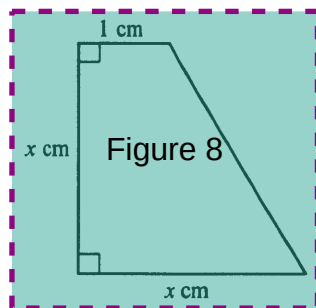
- Label 7 In the figure, a square of side  $x$  cm is cut into 9 equal squares. If the total perimeter of the 9 small squares is 72 cm more than the perimeter of the original square, then  $x =$



- A. 6
- B. 8
- C. 9
- D. 12
- E. 18

- Label 8 The figure shows a trapezium of area  $6 \text{ cm}^2$ . Find  $x$

- A. 2
- B. 3
- C. 4
- D.  $\sqrt{6}$
- E.  $\sqrt{11}$



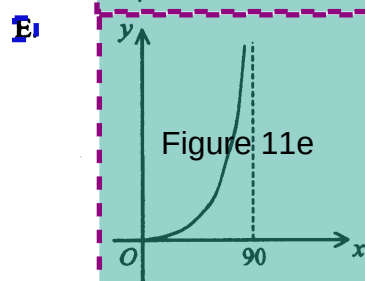
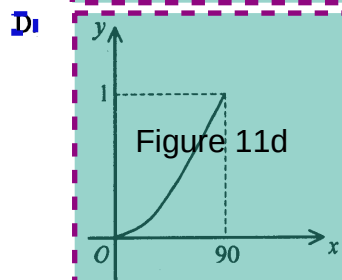
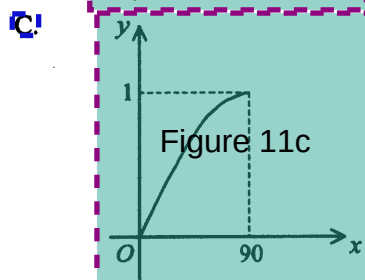
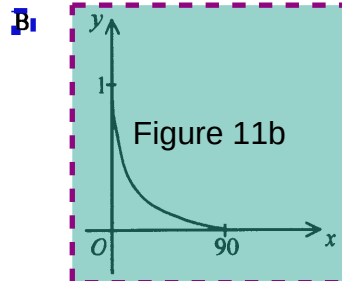
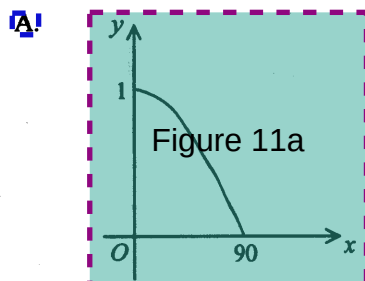
- Label 9 Let  $f(x) = x^3 - 2x^2 - 5x + 6$ . It is known that  $f(1) = 0$ .  $f(x)$  can be factorized as

- A.  $(x-1)^2(x+6)$
- B.  $(x-1)(x+1)(x+6)$
- C.  $(x-1)(x-2)(x+3)$
- D.  $(x-1)(x+2)(x-3)$
- E.  $(x+1)(x-2)(x-3)$

- Label 10 If  $3x^2 + ax + 7 = 3(x-2)^2 + b$ , then

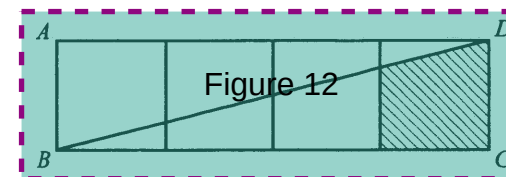
- A.  $a = -12, b = -5$
- B.  $a = -12, b = 7$
- C.  $a = -4, b = 3$
- D.  $a = 0, b = -5$
- E.  $a = 0, b = 19$

Label 11 Which of the following may represent the graph of  $y = \tan x^\circ$  for  $0 \leq x \leq 90$ ?

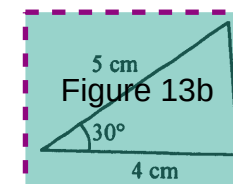
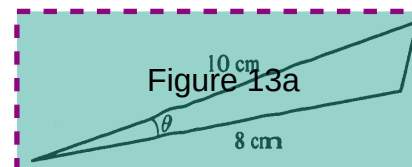


Label 12 In the figure,  $ABCD$  is a rectangle formed by four squares each of area  $1 \text{ cm}^2$ .  $DB$  is a diagonal. Find the area of the shaded region.

- A.  $\frac{9}{10} \text{ cm}^2$
- B.  $\frac{7}{8} \text{ cm}^2$
- C.  $\frac{5}{6} \text{ cm}^2$
- D.  $\frac{4}{5} \text{ cm}^2$
- E.  $\frac{3}{4} \text{ cm}^2$



Label 13 In the figure, the areas of the two triangles are equal. Find  $\theta$ .



- A.  $7.2^\circ$  (correct to the nearest  $0.1^\circ$ )
- B.  $7.5^\circ$  (correct to the nearest  $0.1^\circ$ )
- C.  $14.5^\circ$  (correct to the nearest  $0.1^\circ$ )
- D.  $15^\circ$
- E.  $30^\circ$

Label 14 A man bought two books at \$30 and \$70 respectively. He sold the first one at a profit of 20% and the second one at a loss of 10%. On the whole, he

- A. lost 1%
- B. lost 10%
- C. gained 1%
- D. gained 10%
- E. gained 13%

15. The 1st and 10th terms of an arithmetic sequence are 2 and 29 respectively. The 20th term of the sequence is

- A. 56
- B. 58
- C. 59
- D. 60
- E. 62

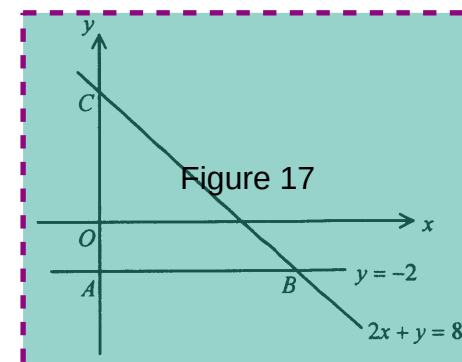
Label 16 Which of the following could be a geometric sequence/geometric sequences?

- I.  $3, 3^3, 3^5, 3^7, \dots$
- II.  $9, 99, 999, 9999, \dots$
- III.  $10, -100, 1000, -10000, \dots$

- A. III only
- B. I and II only
- C. I and III only
- D. I and II only
- E. I, II and III

Label 17 In the figure, find the area of  $\triangle ABC$ .

- A. 1
- B. 3
- C. 6
- D. 20
- E. 25



Label 18 Consider the three straight lines

$$L_1 : 6x + 4y - 3 = 0 ,$$

$$L_2 : y = -\frac{3}{2}x + 4 \text{ and}$$

$$L_3 : 6x - 4y + 3 = 0 .$$

Which of the following is/are true?

I.  $L_1 \parallel L_2$

II.  $L_2 \parallel L_3$

III.  $L_1 \perp L_3$

A. I only

B. II only

C. III only

D. I and III only

E. II and III only

Label 19

Label 19 In the figure,  $ABCD$  is a parallelogram. Find  $\angle BDE$ !

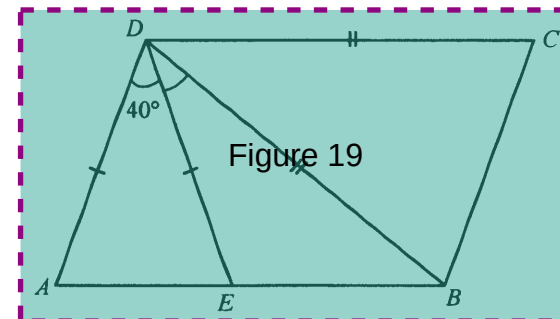
A.  $30^\circ$

B.  $35^\circ$

C.  $40^\circ$

D.  $50^\circ$

E.  $55^\circ$



Label 20 In the figure,  $O$  is the centre of the circle.  $EAOB$  and  $EDC$  are straight lines. Find  $x$ .

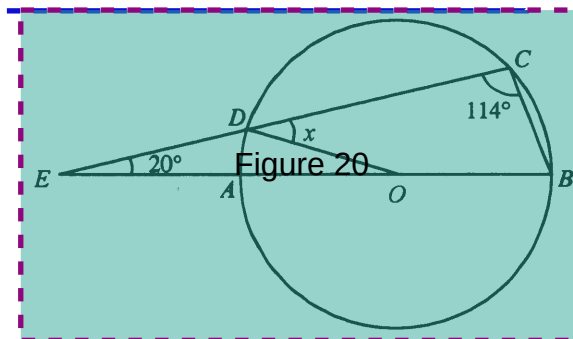
A.  $40^\circ$

B.  $46^\circ$

C.  $57^\circ$

D.  $66^\circ$

E.  $68^\circ$



Label 21 Two fair dice are thrown. Find the probability that at least one "6" occurs.

- A.  $\frac{1}{3}$
- B.  $\frac{1}{6}$
- C.  $\frac{5}{18}$
- D.  $\frac{7}{36}$
- E.  $\frac{11}{36}$

Label 22 A bag contains six balls which are marked with the numbers  $-3$ ,  $-2$ ,  $-1$ ,  $1$ ,  $2$  and  $3$  respectively. Two balls are drawn randomly from the bag. Find the probability that the sum of the numbers drawn is zero.

- A.  $\frac{1}{30}$
- B.  $\frac{1}{10}$
- C.  $\frac{1}{5}$
- D.  $\frac{1}{3}$
- E.  $\frac{1}{2}$

Label 23  $\{x, x+2, x+4, x+6, x+8\}$  and  $\{x+1, x+3, x+5, x+7, x+9\}$  are two groups of numbers. Which of the following is/are true?

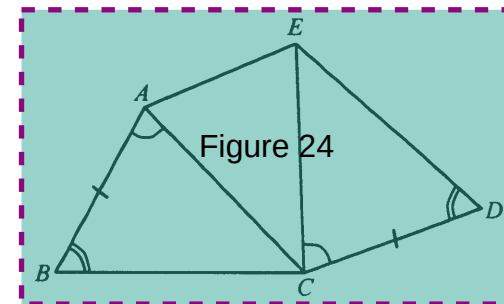
- I. The two groups of numbers have the same range.
- II. The two groups of numbers have the same standard deviation.
- III. The two groups of numbers have the same mean.

- A. I only
- B. II only
- C. III only
- D. I and II only
- E. I and III only

Label 24 In the figure,  $AB = CD$ ,  $\angle CAB = \angle ECD$  and  $\angle ABC = \angle CDE$ . Which of the following must be true?

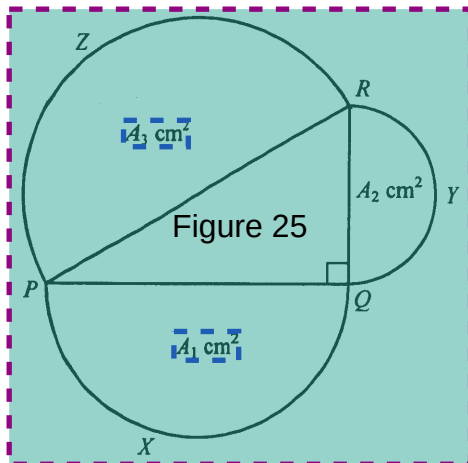
- I.  $\triangle ABC \cong \triangle CDE$
- II.  $\triangle ABC \sim \triangle EAC$
- III.  $EAC$  is an isosceles triangle

- A. I only
- B. III only
- C. I and II only
- D. I and III only
- E. I, II and III



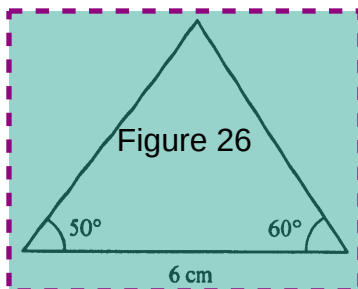
Label 25 In the figure,  $PXQ$ ,  $QYR$  and  $RZP$  are semicircles with areas  $A_1 \text{ cm}^2$ ,  $A_2 \text{ cm}^2$  and  $A_3 \text{ cm}^2$  respectively. If  $A_1 = 12$  and  $A_2 = 5$ , find  $A_3$ .

- A. 3
- B. 7
- C. 169
- D.  $13\pi$
- E.  $\frac{169}{8}\pi$



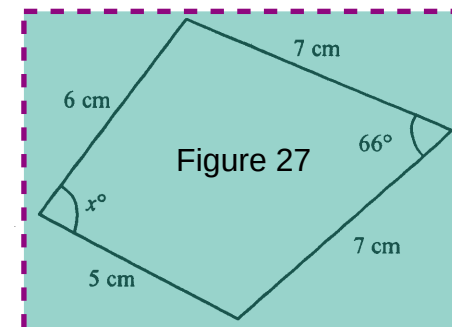
Label 26 In the figure, find the area of the triangle correct to the nearest  $0.1 \text{ cm}^2$ .

- A.  $7.3 \text{ cm}^2$
- B.  $10.7 \text{ cm}^2$
- C.  $12.7 \text{ cm}^2$
- D.  $15.0 \text{ cm}^2$
- E.  $19.1 \text{ cm}^2$

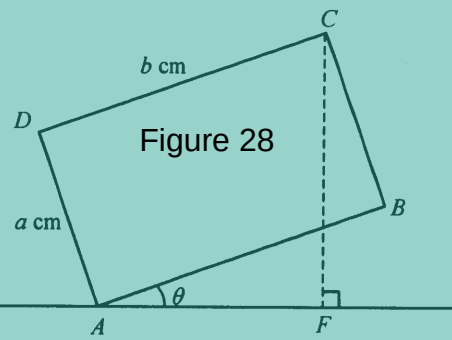


Label 27 In the figure, find  $x$  correct to 3 significant figures!

- A. 63.8
- B. 78.5
- C. 84.5
- D. 87.3
- E. 89.1



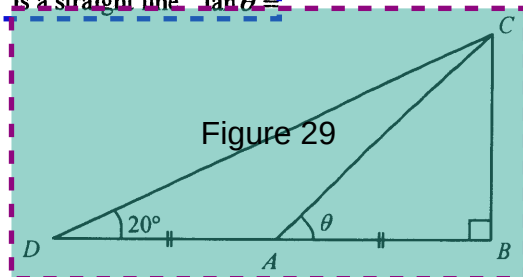
Label 28 In the figure,  $ABCD$  is a rectangle. Find  $CF$ .



- A.  $(a+b) \sin \theta \text{ cm}$
- B.  $(a+b) \cos \theta \text{ cm}$
- C.  $(a \sin \theta + b \cos \theta) \text{ cm}$
- D.  $(a \cos \theta + b \sin \theta) \text{ cm}$
- E.  $\sqrt{a^2 + b^2} \sin 2\theta \text{ cm}$



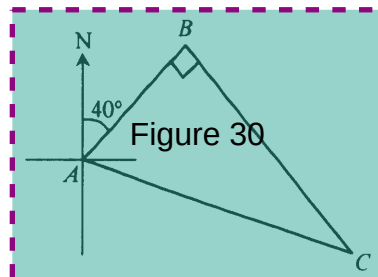
Label 29 In the figure,  $DAB$  is a straight line.  $\tan \theta =$



- A.  $2 \tan 20^\circ$
- B.  $\frac{1}{2} \tan 20^\circ$
- C.  $\frac{2}{\tan 20^\circ}$
- D.  $\frac{1}{2 \tan 20^\circ}$
- E.  $\tan 40^\circ$

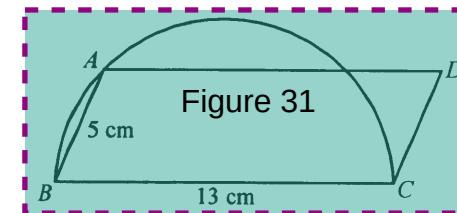
Label 30 According to the figure, the bearing of  $B$  from  $C$  is

- A.  $050^\circ$
- B.  $130^\circ$
- C.  $140^\circ$
- D.  $310^\circ$
- E.  $320^\circ$

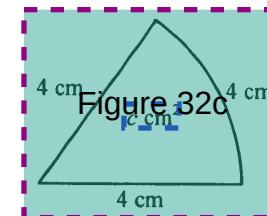
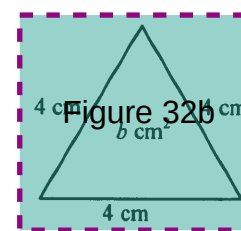
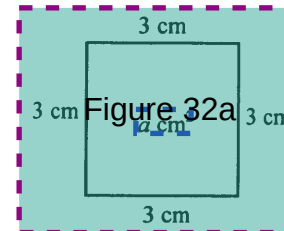


Label 31 In the figure,  $CAB$  is a semicircle and  $ABCD$  is a parallelogram. Find the area of  $ABCD$ .

- A.  $65 \text{ cm}^2$
- B.  $60 \text{ cm}^2$
- C.  $52 \text{ cm}^2$
- D.  $32.5 \text{ cm}^2$
- E.  $30 \text{ cm}^2$



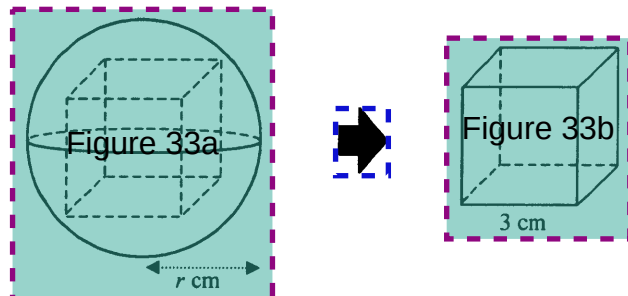
Label 32 The figure shows a square, a triangle and a sector with areas  $a \text{ cm}^2$ ,  $b \text{ cm}^2$  and  $c \text{ cm}^2$  respectively.



Which of the following is true?

- A.  $a > b > c$
- B.  $a > c > b$
- C.  $b > a > c$
- D.  $b > c > a$
- E.  $c > a > b$

Label 33 In the figure, a solid wooden sphere of radius  $r$  cm is to be cut into a cube of side 3 cm. Find the smallest possible value of  $r$ .



A.  $\frac{3\sqrt{3}}{2}$

B.  $\frac{3\sqrt{2}}{2}$

C.  $\frac{3}{2}$

D.  $3\sqrt{3}$

E.  $3\sqrt{2}$

Label 34 If  $9a^2 - b^2 = 0$  and  $ab < 0$ , then  $\frac{a-b}{a+b} =$

A.  $\frac{1}{2}$

B.  $\frac{1}{2}$

C.  $0$

D.  $\frac{1}{2}$

E.  $0$

Label 35  $y$  varies directly as  $x^2$  and inversely as  $\sqrt{z}$ . If  $y=1$  when  $x=2$  and  $z=9$ , find  $y$  when  $x=1$  and  $z=4$ .

A.  $\frac{2}{3}$

B.  $\frac{8}{3}$

C.  $\frac{1}{6}$

D.  $\frac{3}{8}$

E.  $\frac{9}{26}$

Label 36 Tea  $A$  and tea  $B$  are mixed in the ratio  $x : y$  by weight.  $A$  costs \$80/kg and  $B$  costs \$100/kg. If the cost of  $A$  is increased by 10% and that of  $B$  is decreased by 12%, the cost of the mixture per kg remains unchanged. Find  $x : y$ .

A.  $1:1$

B.  $2:3$

C.  $3:2$

D.  $5:6$

E.  $6:5$

Section B

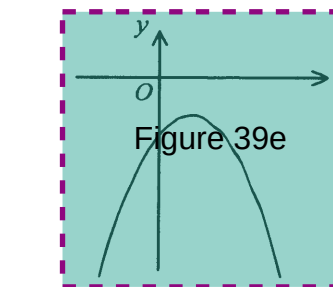
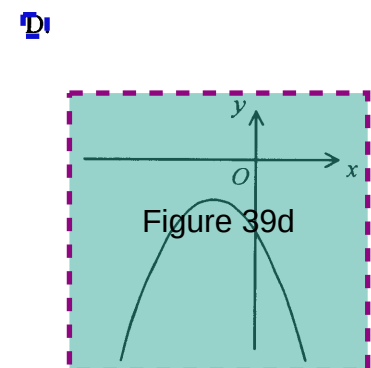
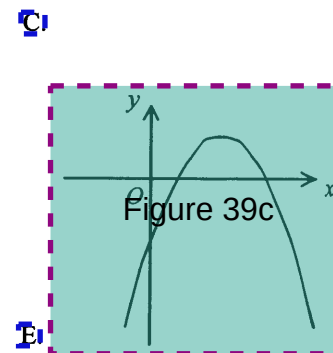
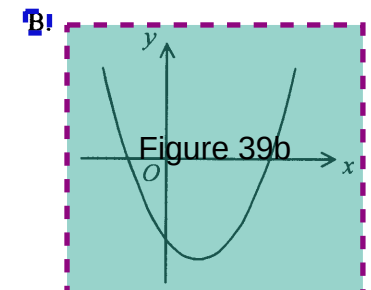
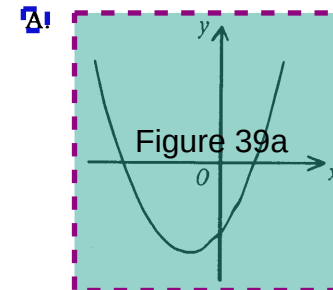
Label 37 Simplify  $\frac{a+b}{a+b} + \frac{2ab}{b-a} - \frac{2ab}{a^2-b^2}$ .

- A.  $\frac{a+b}{a-b}$
- B.  $\frac{a-b}{a+b}$
- C.  $\frac{-a^2+b^2+4ab}{a^2-b^2}$
- D.  $\frac{a^2+b^2}{a^2-b^2}$
- E. I

Label 38 If  $\log(x-a)=3$ , then  $x=$

- A.  $10^{3+a}$
- B.  $a^3$
- C.  $1000a$
- D.  $1000+a$
- E.  $30+a$

Label 39 Which of the following may represent the graph of  $y = -x^2 + 2x - 3$ ?



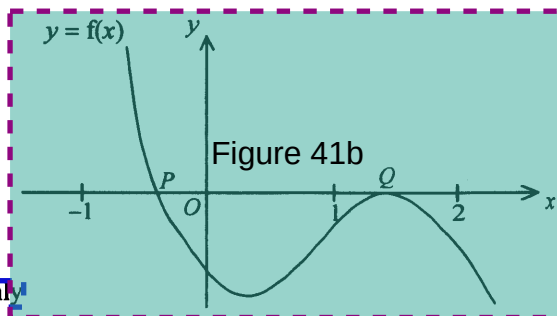
Label 40 If  $\left(\frac{\sqrt{5}}{2} + 1\right)x = \sqrt{2}$ , then  $x =$

- A.  $2\sqrt{10} - 2$   
 B.  $2\sqrt{10} - 4\sqrt{2}$   
 C.  $2\sqrt{10} + 4\sqrt{2}$   
 D.  $\frac{\sqrt{10}}{2}$   
 E.  $\frac{\sqrt{10}}{3}$

Label 41 In the figure, the graph of  $y = f(x)$  intersects the  $x$ -axis at  $P$  and  $Q$  only. In order to find a root of  $f(x) = 0$  using the method of bisection, which of the following intervals can you start with?

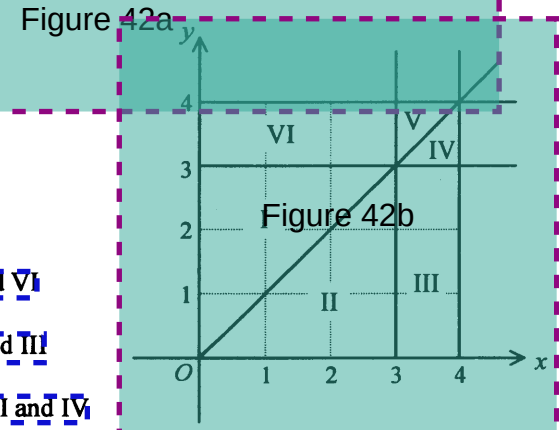
- I.  $-1 < x < 0$   
 II.  $0 < x < 1$   
 III.  $1 < x < 2$

- A. I only  
 B. III only  
 C. I and II only  
 D. I and III only  
 E. I, II and III



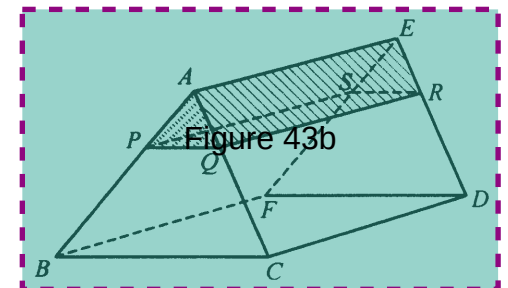
Label 42 According to the figure, which of the following represents the solution of  $\begin{cases} 0 \leq x \leq 4 \\ x \geq y \\ 0 \leq y \leq 3 \end{cases}$ ?

- A. Region I  
 B. Region II  
 C. Regions I and VI  
 D. Regions II and III  
 E. Regions II, III and IV



Label 43 In the figure,  $ABCDEF$  is a right triangular prism. It is cut into two parts along the plane  $PQRS$ , which is parallel to the face  $BCDF$ , and volume of the prism  $APQRES$ .  $AP : PB = 2 : 5$ . Find  $\frac{\text{volume of the prism } APQRES}{\text{volume of the prism } ABCDEF}$ .

- A.  $\frac{2}{7}$   
 B.  $\frac{4}{25}$   
 C.  $\frac{4}{49}$   
 D.  $\frac{8}{125}$   
 E.  $\frac{8}{343}$

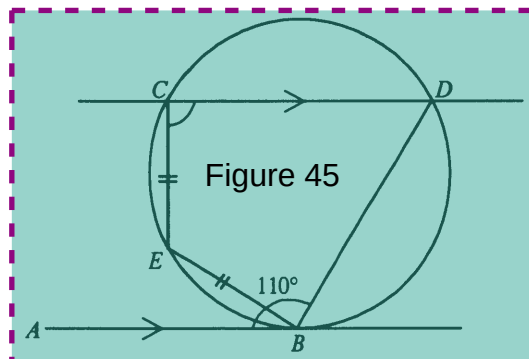


Label 44  $\pi$  degrees =

- A  $\frac{\pi^2}{180}$  radian
- B  $\frac{180}{\pi^2}$  radians
- C  $\frac{\pi}{180}$  radian
- D  $180$  radians
- E  $1$  radian

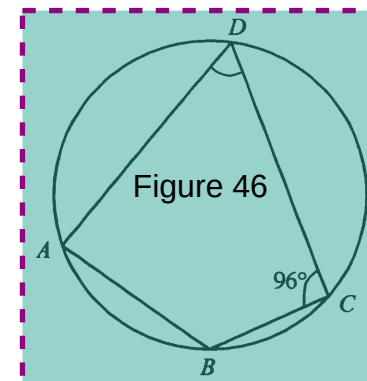
Label 45 In the figure,  $AB$  is tangent to the circle at  $B$ . Find  $\angle DCE$ .

- A  $70^\circ$
- B  $75^\circ$
- C  $90^\circ$
- D  $95^\circ$
- E  $105^\circ$



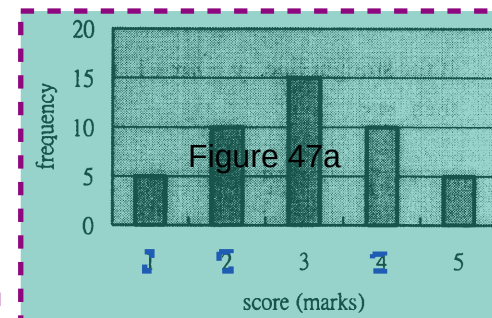
Label 46 In the figure,  $\widehat{AB} : \widehat{BC} : \widehat{CD} = 2 : 1 : 3$ . Find  $\angle ADC$ .

- A  $56^\circ$
- B  $60^\circ$
- C  $63^\circ$
- D  $72^\circ$
- E  $84^\circ$



Label 47 The bar chart below shows the distribution of scores of a test. Find the mean deviation of the scores of the test.

- A  $0$  mark
- B  $\frac{8}{9}$  mark
- C  $\frac{\sqrt{5}}{3}$  mark
- D  $\frac{\sqrt{5}}{3}$  marks
- E  $\frac{6}{5}$  marks



Label 48 If the centre of the circle  $x^2 + y^2 + kx + (k+1)y - 3 = 0$  lies on  $x + y + 1 = 0$ , find  $k$ .

- A.  $\frac{3}{2}$
- B.  $\frac{1}{2}$
- C. 0
- D. -1
- E.  $-\frac{3}{2}$

Label 49 If the straight line  $y = mx + 1$  is tangent to the circle  $(x-2)^2 + y^2 = 1$ , then  $m =$

- A.  $-\frac{4}{3}$
- B. 0
- C.  $\frac{4}{3}$
- D. 0 or  $-\frac{4}{3}$
- E. 0 or  $\frac{4}{3}$

Label 50  $A(-1, -4)$  and  $B(3, 4)$  are two points. The line  $x - y = 0$  cuts  $AB$  at  $P$  so that  $AP : PB = r : 1$ . Find  $r$ .

- A. 3
- B. 2
- C. 1
- D.  $\frac{1}{2}$
- E.  $\frac{1}{3}$

Label 51 If  $\cos \theta = \frac{1}{k}$  and  $0^\circ < \theta < 90^\circ$ , then  $\tan(\theta - 270^\circ) =$

- A.  $\frac{k}{\sqrt{1-k^2}}$
- B.  $\frac{1}{\sqrt{k^2-1}}$
- C.  $\frac{1}{\sqrt{k^2-1}}$
- D.  $-\sqrt{k^2-1}$
- E.  $\sqrt{k^2-1}$

Label 48 If the centre of the circle  $x^2 + y^2 + kx + (k+1)y - 3 = 0$  lies on  $x + y + 1 = 0$ , find  $k$ .

- A.  $\frac{3}{2}$
- B.  $\frac{1}{2}$
- C. 0
- D. -1
- E.  $-\frac{3}{2}$

Label 49 If the straight line  $y = mx + 1$  is tangent to the circle  $(x - 2)^2 + y^2 = 1$ , then  $m =$

- A.  $-\frac{4}{3}$
- B. 0
- C.  $\frac{4}{3}$
- D. 0 or  $-\frac{4}{3}$
- E. 0 or  $\frac{4}{3}$

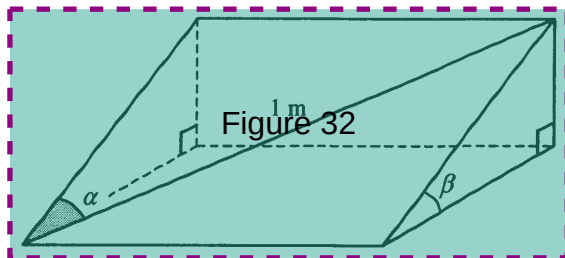
Label 50  $A(-1, -4)$  and  $B(3, 4)$  are two points. The line  $x - y = 0$  cuts  $AB$  at  $P$  so that  $AP : PB = r : 1$ . Find  $r$ .

- A. 3
- B. 2
- C. 1
- D.  $\frac{1}{2}$
- E.  $\frac{1}{3}$

Label 51 If  $\cos \theta = \frac{1}{k}$  and  $0^\circ < \theta < 90^\circ$ , then  $\tan(\theta - 270^\circ) =$

- A.  $\frac{k}{\sqrt{1-k^2}}$
- B.  $\frac{1}{\sqrt{k^2-1}}$
- C.  $\frac{1}{\sqrt{k^2-1}}$
- D.  $-\sqrt{k^2-1}$
- E.  $\sqrt{k^2-1}$

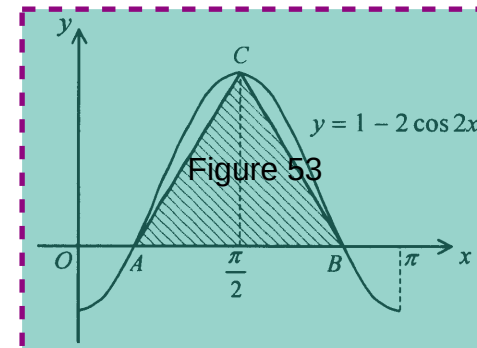
Label 32 The figure shows a right triangular prism. Find its volume.



- A.  $\frac{1}{3} \sin^2 \alpha \cos \alpha \sin \beta \cos \beta \text{ m}^3$
- B.  $\frac{1}{3} \sin \alpha \cos^2 \alpha \sin \beta \cos \beta \text{ m}^3$
- C.  $\frac{1}{2} \sin \alpha \cos \alpha \sin \beta \cos \beta \text{ m}^3$
- D.  $\frac{1}{2} \sin^2 \alpha \cos \alpha \sin \beta \cos \beta \text{ m}^3$
- E.  $\frac{1}{2} \sin \alpha \cos^2 \alpha \sin \beta \cos \beta \text{ m}^3$

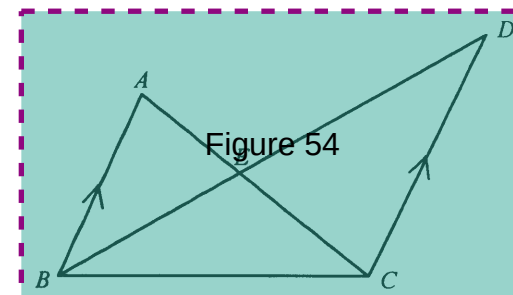
Label 53 In the figure, the area of  $\triangle ABC$  is

- A.  $\frac{\pi}{3}$
- B.  $\frac{2\pi}{3}$
- C.  $\pi$
- D.  $\frac{4\pi}{3}$
- E.  $2\pi$



Label 54 In the figure,  $AEC$  and  $BED$  are straight lines. If the area of  $\triangle ABE = 4 \text{ cm}^2$  and the area of  $\triangle BCE = 5 \text{ cm}^2$ , find the area of  $\triangle CDE$ .

- A.  $4.5 \text{ cm}^2$
- B.  $5 \text{ cm}^2$
- C.  $6 \text{ cm}^2$
- D.  $6.25 \text{ cm}^2$
- E.  $9 \text{ cm}^2$



END OF PAPER